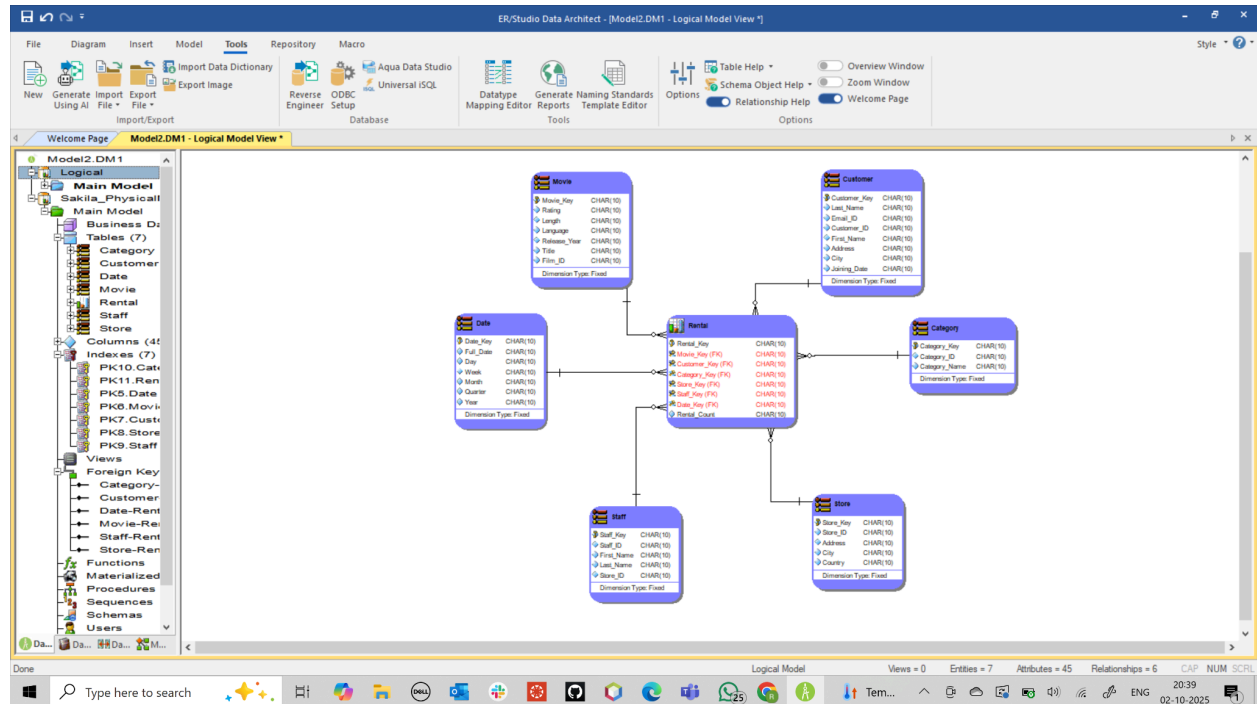


Dimensional Modeling

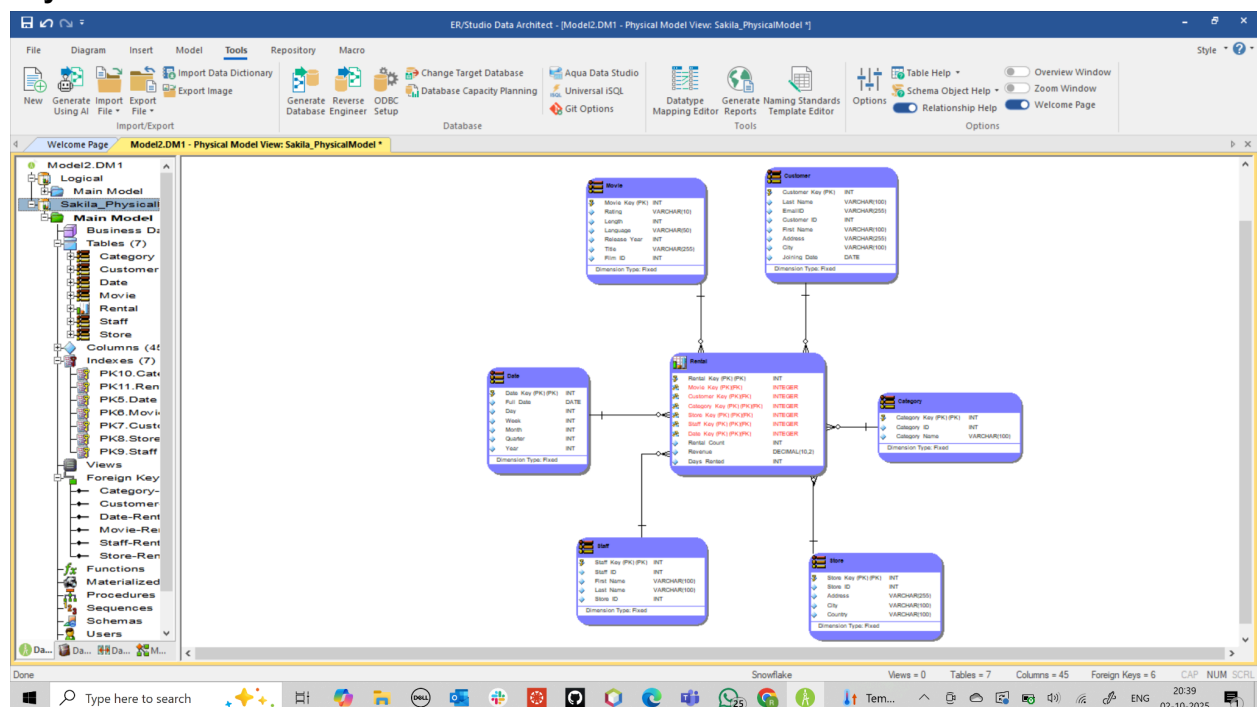
Name: Riya Patel

NUID: 002059816

Logical Model



Physical Model



Changed DataTypes

The screenshot shows the ER/Studio Data Architect interface with the 'Movie' table selected in the 'Table Editor'. The table's columns and their data types are as follows:

Column	Domain	Datatype	Nulls
1 Movie_Key		INT	NOT NULL
2 Rating		VARCHAR(10)	NULL
3 Length		INT	NULL
4 Language		VARCHAR(50)	NULL
5 Release_Year		INT	NULL
6 Title		VARCHAR(255)	NULL
7 Film_ID		INT	NULL

The left sidebar shows the 'Main Model' tree with 'Sakila_Physical' selected. The bottom status bar indicates 'Views = 0', 'Tables = 7', 'Columns = 45', and 'Foreign Keys = 6'.

The screenshot shows the ER/Studio Data Architect interface with the 'Customer' table selected in the 'Table Editor'. The table's columns and their data types are as follows:

Column	Domain	Datatype	Nulls
1 Customer_Key		INT	NOT NULL
2 Last_Name		VARCHAR(100)	NULL
3 EmailID		VARCHAR(255)	NULL
4 Customer_ID		INT	NULL
5 First_Name		VARCHAR(100)	NULL
6 Address		VARCHAR(255)	NULL
7 City		VARCHAR(100)	NULL
8 Joining_Date		DATE	NULL

The left sidebar shows the 'Main Model' tree with 'Sakila_Physical' selected. The bottom status bar indicates 'Views = 0', 'Tables = 7', 'Columns = 45', and 'Foreign Keys = 6'.

ER/Studio Data Architect - [Model2.DM1 - Physical Model View: Sakila_PhysicalModel1]

File Diagram Insert Model Tools Repository Macro

Generate Import Export Using AI File Import/Export

Generate Reverse Database Engineer ODBC Setup Database Capacity Planning Universal ISQL Datatype Mapping Editor Reports Template Editor

Table Editor

Entity Name: Category Table Name: Category Owner: [dropdown]

Physical Only Do Not Generate

Permissions PreSQL & PostSQL Naming Standards Compare Options Data Lineage Security Information Attachment Bindings

Note Where Used Storage Clustering Constraints Dependencies Capacity Planning

Column	Domain	Datatype	Nulls
1 Category_Key (PK)	INT	INT	NOT NULL
2 Category_ID	INT	INT	NULL
3 Category_Name	VARCHAR(100)	VARCHAR(100)	NULL

Add Edit Delete Up Down OK Cancel Help

Category

- Category_Key (PK) INT
- Category_ID INT
- Category_Name VARCHAR(100)

Dimension Type: Fixed

Done

Type here to search

Snowflake Views = 0 Tables = 7 Columns = 45 Foreign Keys = 6 CAP NUM SCRL

20:41 02-10-2025

ER/Studio Data Architect - [Model2.DM1 - Physical Model View: Sakila_PhysicalModel1]

File Diagram Insert Model Tools Repository Macro

Generate Import Export Using AI File Import/Export

Generate Reverse Database Engineer ODBC Setup Database Capacity Planning Universal ISQL Datatype Mapping Editor Reports Template Editor

Table Editor

Entity Name: Store Table Name: Store Owner: [dropdown]

Physical Only Do Not Generate

Permissions PreSQL & PostSQL Naming Standards Compare Options Data Lineage Security Information Attachment Bindings

Note Where Used Storage Clustering Constraints Dependencies Capacity Planning

Column	Domain	Datatype	Nulls
1 Store_Key (PK)	INT	INT	NOT NULL
2 Store_ID	INT	INT	NULL
3 Address	VARCHAR(255)	VARCHAR(255)	NULL
4 City	VARCHAR(100)	VARCHAR(100)	NULL
5 Country	VARCHAR(100)	VARCHAR(100)	NULL

Add Edit Delete Up Down OK Cancel Help

Store

- Store_Key (PK) INT
- Store_ID INT
- Address VARCHAR(255)
- City VARCHAR(100)
- Country VARCHAR(100)

Dimension Type: Fixed

Done

Type here to search

Snowflake Views = 0 Tables = 7 Columns = 45 Foreign Keys = 6 CAP NUM SCRL

20:42 02-10-2025

ER/Studio Data Architect - [Model2.DM1 - Physical Model View: Sakila_PhysicalModel1]

File Diagram Insert Model Tools Repository Macro

Generate Import Export Using AI File Import/Export

Generate Reverse Database Engineer ODBC Setup Database Capacity Planning Aqua Data Studio Universal ISQL Datatype Mapping Editor Reports Template Editor

Table Editor

Entity No: Staff Table Name: Staff Owner: [dropdown]

Physical Only Do Not Generate

Permissions PreSQL & PostSQL Naming Standards Compare Options Data Lineage Security Information Attachment Bindings Note Where Used Storage Clustering Constraints Dependencies Capacity Planning

Column	Domain	Datatype	Nulls
1 Staff_Key (PK)	INT	INT	NOT NULL
2 Staff_ID	INT	INT	NULL
3 First_Name	VARCHAR(100)	VARCHAR(100)	NULL
4 Last_Name	VARCHAR(100)	VARCHAR(100)	NULL
5 Store_ID	INT	INT	NULL

Category

- Category_Key (PK) (PK) INT
- Category_ID INT
- Category_Name VARCHAR(100)

Dimension Type: Fixed

Store_Key (PK) (PK) INT

- Store_ID INT
- Address VARCHAR(255)
- City VARCHAR(100)
- Country VARCHAR(100)

Done

Type here to search

Snowflake Views = 0 Tables = 7 Columns = 45 Foreign Keys = 6 CAP NUM SCRL

20:42 02-10-2025

ER/Studio Data Architect - [Model2.DM1 - Physical Model View: Sakila_PhysicalModel1]

File Diagram Insert Model Tools Repository Macro

Generate Import Export Using AI File Import/Export

Generate Reverse Database Engineer ODBC Setup Database Capacity Planning Aqua Data Studio Universal ISQL Datatype Mapping Editor Reports Template Editor

Table Editor

Entity No: Date Table Name: Date Owner: [dropdown]

Physical Only Do Not Generate

Permissions PreSQL & PostSQL Naming Standards Compare Options Data Lineage Security Information Attachment Bindings Note Where Used Storage Clustering Constraints Dependencies Capacity Planning

Column	Domain	Datatype	Nulls
1 Date_Key (PK)	INT	INT	NOT NULL
2 Full_Date	DATE	DATE	NULL
3 Day	INT	INT	NULL
4 Week	INT	INT	NULL
5 Month	INT	INT	NULL
6 Quarter	INT	INT	NULL
7 Year	INT	INT	NULL

Category

- Category_Key (PK) (PK) INT
- Category_ID INT
- Category_Name VARCHAR(100)

Dimension Type: Fixed

Store_Key (PK) (PK) INT

- Store_ID INT
- Address VARCHAR(255)
- City VARCHAR(100)
- Country VARCHAR(100)

Done

Type here to search

Snowflake Views = 0 Tables = 7 Columns = 45 Foreign Keys = 6 CAP NUM SCRL

20:42 02-10-2025

SQL script in MEDIA_USER database

The screenshot shows the Snowflake web interface with a workspace named 'My Workspace' containing a file 'Untitled.sql'. The SQL script in the editor is:

```
167 FOREIGN KEY ("Date_Key (PK)")
168 REFERENCES Date("Date_Key (PK)")
169 ;
170
171 ALTER TABLE Rental ADD CONSTRAINT RefMovie11
172 FOREIGN KEY (Movie_Key)
173 REFERENCES Movie(Movie_Key)
174 ;
175
```

The 'Results' pane shows a single row with the status 'Statement executed successfully.' The 'Query History' pane shows the current query and three previous queries, all executed successfully within 1 minute.

Database Explorer shows the following structure:

- MEDIA_DB
- SNOWFLAKE
- SNOWFLAKE_LEARNING_DB
- SNOWFLAKE_SAMPLE_DATA
- USER\$MEDIA_USER

The screenshot shows the Snowflake web interface with a workspace named 'My Workspace' containing a file 'DDL_MediaUser.sql'. The SQL script in the editor is:

```
169 ;
170
171 ALTER TABLE Rental ADD CONSTRAINT RefMovie11
172 FOREIGN KEY (Movie_Key)
173 REFERENCES Movie(Movie_Key)
174 ;
175
176 SHOW TABLES IN SCHEMA MEDIA_DB.SAKILA;
177
178
179
```

The 'Results' pane shows 7 rows of data, including columns for 'created_on', 'name', 'database_name', 'schema_name', 'kind', 'comment', 'cluster_by', and '# rows'. The 'Query History' pane shows the current query and three previous queries, all executed successfully within 1 minute.

Database Explorer shows the following structure:

- MEDIA_DB
- INFORMATION_SCHEMA
- SAKILA
 - Tables
 - CATEGORY
 - CUSTOMER
 - DATE
 - MOVIE
 - RENTAL
 - STAFF
 - STORE
- SNOWFLAKE
- SNOWFLAKE_LEARNING_DB
- SNOWFLAKE_SAMPLE_DATA
- USER\$MEDIA_USER